

Class Section: _____

Assignment #2

MG-2009 Data Analysis for Business -II

Sections: BS(Fin-Tech) All Sections

Submission Date: 06/04/2026

Total Marks: 105 [100 Assignment Marks + 5 Presentation Marks]

Instructions:

- Assignment should be done on A4 size paper.
- There is no tolerance for plagiarism.
- Write down your class section on top right corner of the title page.

Q#1: State the Null & Alternative for the following situations: **[8 × 1 = 8 marks]**

- A company claims that a new advertising campaign increases sales by 15%. Test whether the claim is true.
- A factory states that the average weight of its packaged product is 500 grams. Test whether the actual mean weight differs from this value.
- A manager believes that employees complete tasks in less than 8 hours on average. Test this assumption.
- A survey claims that 70% of customers are satisfied with a service. Test whether the satisfaction rate is different from 70%.
- A logistics company claims that the average delivery time is 2 days. Test whether the actual delivery time is greater than 2 days.
- An e-commerce business claims that at least 5% of visitors make a purchase. Test whether the conversion rate is less than 5%.
- A bank reports that 40% of loan applications are approved. Test whether the approval rate is higher than 40%.
- A manufacturer claims that a new machine produces more than 100 units per hour on average. Test this claim.

Q#2: State Type-I & Type-II error for the following claims. **[8 × 1 = 8 marks]**

- I. A pharmaceutical company tests whether a new drug lowers blood pressure.
- II. A factory tests whether a batch of products is defective.
- III. A bank uses a system to detect fraudulent transactions.
- IV. A company tests whether an employee meets performance standards.
- V. An online store tests whether a new design increases conversion rates.

- VI. A company tests whether a candidate is suitable for a job.
- VII. A health department tests whether food at a restaurant is contaminated.
- VIII. A school tests whether a new teaching method improves student performance.

Q#3: Draw the critical region.

[8 × 1 = 8 marks]

- I. A company claims that the average battery life of its product is more than 10 hours.
- II. A researcher wants to check if the average income of workers is different from \$30,000.
- III. A factory manager believes that the defect rate is less than 3%.
- IV. A school claims that the average test score is equal to 75 marks.
- V. A bank wants to test whether the average loan processing time is greater than 5 days.
- VI. An online store checks if the return rate is not equal to 8%.
- VII. A nutritionist claims that the average calorie intake is less than 2,000 calories.
- VIII. A company tests whether the average salary of employees is different from \$50,000.

Q#4: A financial analyst is comparing the average monthly profit (in thousands of dollars) generated by two different branches of a company. Two independent random samples were taken from each branch:

- Branch A ($n_1 = 6$): 12.5, 15.2, 9.8, 11.3, 8.7, 13.1
- Branch B ($n_2 = 5$): 7.4, 6.8, 10.2, 14.5, 9.1

Assuming that the monthly profits for both branches are normally distributed with unequal variations, does the data provide sufficient evidence to suggest that the difference in average monthly profits between Branch A and Branch B is at least 4 thousand dollars? Estimate the population difference using 98% confidence. **[12 + 8 = 20 marks]**

Q#5: Uber-Eats introduced the fastest delivery system and claimed that it serves its food on average 35.5 minutes with standard deviation 1.7 minutes. In order to know whether the Uber-Eats claim was right, strongest competitor Door Dash made 40 orders obtaining average time of 36.1 minutes for the order to be delivered.

- a) Test the hypothesis whether the claim of Uber-Eats is valid or not.
- b) Calculate 99% confidence interval for the population mean “ μ ” and interpret the results.

[10 + 5 = 15 Marks]

Q#6: Based on an indication that mean daily car-rental rates may be higher in Melbourne than in Sydney, a survey of eight car-rental companies in Melbourne is taken and the sample mean car-rental rate is \$47 with a standard deviation of \$3. Further, a survey of nine car-rental companies in Sydney results in a sample mean of \$44 and a

standard deviation of \$3. Use $\alpha = 0.01$ to test whether the average daily car-rental rates in Melbourne are significantly higher than those in Sydney. Assume car-rental rates are normally distributed and the population variances are equal. **[13 Marks]**

Q#7: A computer science teacher wanted to investigate whether AI-based coding instruction could improve student performance compared to traditional textbook-based learning. She recorded the grades of ten students before and after implementing AI-assisted teaching. Before the midterm, while using textbooks, the students' scores were: 68, 72, 64, 70, 66, 69, 71, 65, 67, and 73. After the midterm, using AI-based coding tools, the students' scores were recorded as: 78, 82, 75, 80, 73, 77, 79, 74, 76, and 81. Using a significance level of 0.01, test whether the AI-based teaching method had a significant effect on student grades. **[13 Marks]**

Q#8:

- a) Companies that recently developed new products were asked to rate which activities were most difficult to accomplish with new products. Options included such activities as assessing market potential, market testing, completing the design and developing a business plan. A researcher wants to conduct a similar study to compare the results between two industries: the computer hardware industry and the banking industry. He takes a random sample of 56 computer companies and 89 banks. The researcher asks whether market testing is the most difficult activity to accomplish in developing a new product. Some 48% of the sampled computer companies and 56% of the sampled banks respond that it is the most difficult activity. Use a level of significance of 0.20 to test whether there is a significant difference in the responses to the question between these two industries.
- b) Calculate 98% confidence interval for the difference between population proportion and interpret the results.

[10 + 5 = 15 Marks]